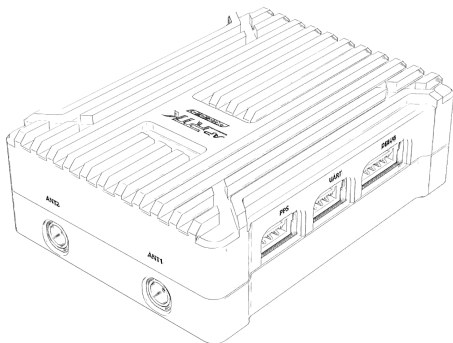


AP-RTK X20D

User Guide



AP-RTK X20D Overview



One u-blox ZED-X20D all-band receiver and two antennas provide RTK position and heading, forwarded with RM3100 compass data to the flight controller over DroneCAN (engineering evaluation).

AP-RTK X20D Specifications

• Basic Specifications

Receiver	u-blox ZED-X20D — single module, dual antenna, HDG 2.00
GNSS Bands	All-band GNSS on both antennas (L1 / L2 / L5 / L6)
MCU	STM32F412Rx, ARM Cortex-M4, 512 KB flash
Compass	PNI RM3100 on I2C3 — axis qualification pending
Heading	Receiver ANT1-to-ANT2 baseline (UBX-NAV-DAHEADING v2) sent as RelPosHeading; the autopilot computes heading with GPS1_MB_OFS as for AP-RTK dual (firmware v1.0.6+)

AP-RTK X20D Specifications

• Basic Specifications

Update Rate	Firmware requests 5 Hz navigation at 230400 baud; end-to-end CAN rate requires measurement
Comm. Protocol	DroneCAN — position, ardupilot.gnss.RelPosHeading (v1.0.6+) and separate magnetometer data
I/O Ports	2× MMCX antenna · CAN · UART (RTCM input) · DEBUG (SWD + console) · PPS/EVENT · USB-C (MCU)
PCB	R3 · 33.00 × 45.17 mm · 4 layers · 0.8 mm
Firmware	AP_Periph · novaX 1.0.6 engineering release · board ID 6205

AP-RTK X20D Specifications

• Basic Specifications

Validation Status
Bench hardware checks passed 2026-09-22 (230400 save, USB and CAN update, heading through the flight controller); outdoor RTK and flight qualification pending

AP-RTK X20D Pinout

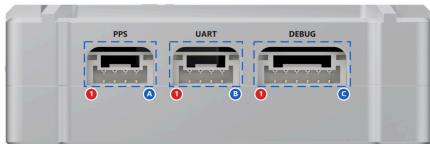
• Pinout (1/3)

AP-RTK X20D Pinout

Each face seen straight on. Each table reads left → right exactly as the picture next to it; number = pin on the R3 board.

A = connector, its table below **1** = pin 1, the red 1 in its table red = power black = GND blue = signal

PPS · UART · DEBUG side



A PPS

4-Pin · JST-GH

1	2	3	4
GND	EVENT	GND	PPS



B UART

4-Pin · JST-GH

1	2	3	4
GND	TX	RX	5V



C DEBUG

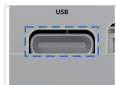
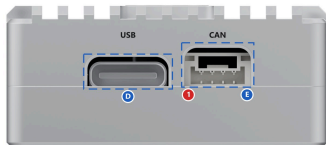
6-Pin · JST-GH

1	2	3	4	5	6
5V	SWDIO	SWCLK	RX	TX	GND

AP-RTK X20D Pinout

• Pinout (2/3)

USB · CAN end



D USB

Type-C

USB 2.0



E CAN

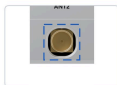
4-Pin · JST-GH

1	2	3	4
GND	CAN_L	CAN_H	5V

AP-RTK X20D Pinout

• Pinout (3/3)

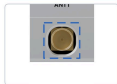
Antenna end



F ANT2

MMCX - slave antenna

RF



G ANT1

MMCX - master antenna

RF

AP-RTK X20D Pinout

• Pinout Notes

Listed from the last pin to pin 1 (the Pinout image shows each connector as seen and marks pin 1): ① UART (5V · RX · TX · GND) · ② CAN (5V · CAN_H · CAN_L · GND) · ③ DEBUG (GND · TX · RX · SWCLK · SWDIO · 5V) · ④ PPS (PPS · GND · EVENT · GND) · ANT1 · ANT2 (MMCX).

AP-RTK X20D Setup

1. **Antenna direction** — ANT1 (RF_IN_1) at the rear, ANT2 (RF_IN_2) at the front, on the centerline at equal height; measure the baseline.
2. **H7E connection** — Connect over DroneCAN and set the parameters below. No X20D-specific H7E firmware build is needed.
3. **Position offset** — Measure ANT1 relative to the centre of gravity and enter GPS1_POS_X/Y/Z (+X forward, +Y right, +Z down).
4. **First-article checks** — On the bench, confirm receiver startup, RTK corrections, position, heading at known directions, reacquisition, RM3100 axes, power and update paths.

AP-RTK X20D Setup

• Flight Controller Parameters

Parameter	Value	Description
CAN_P1_DRIVER	1	Enable the autopilot (H7E) CAN1 port (reboot required after change)
CAN_D1_PROTOCOL	1	DroneCAN protocol on the CAN1 driver
GPS1_TYPE	9	DroneCAN GPS. On firmware older than 4.6 the parameter is GPS_TYPE = 9
GPS_AUTO_CONFIG	2	Automatically configure the DroneCAN GPS
GPS1_MB_TYPE	1	Enable dual-antenna moving baseline (unlocks the offsets below). Pre-4.6: GPS_MB1_TYPE = 1
GPS1_MB_OFS_X	0.50	Master(ANT1) is 0.50 m in front of Slave(ANT2) → +0.50 (positive = Master in front). Set to your measured separation. Pre-4.6: GPS_MB1_OFS_X

AP-RTK X20D Setup

• Flight Controller Parameters

Parameter	Value	Description
GPS1_MB_OFS_Y	0	0 on the centerline (positive = Master to the right of Slave). Pre-4.6: GPS_MB1_OFS_Y
GPS1_MB_OFS_Z	0	0 at equal height (positive = Master below Slave). Pre-4.6: GPS_MB1_OFS_Z
EK3_SRC1_YAW	3	GPS yaw with compass fallback (use 2 for GPS-only yaw)
AHRS_EKF_TYPE	3	Heading works only when AHRS uses EKF3
EK3_ENABLE	1	Enable EKF3
GPS1_POS_X	0.25	Master antenna fore/aft offset from the center of gravity in meters (positive = in front of CoG). Example: with the CoG at the midpoint of the 500 mm baseline, the front Master antenna is half the baseline ahead → +0.25 m. Pre-4.6: GPS_POS1_X

AP-RTK X20D Setup

• Flight Controller Parameters

Parameter	Value	Description
GPS1_POS_Y	0.00	Lateral offset (positive = to the right of CoG). Example: the Master antenna sits on the airframe centerline (Figure 1), so 0. Pre-4.6: GPS_POS1_Y
GPS1_POS_Z	-0.10	Vertical offset (positive = below CoG). Example: the antenna is mounted 0.10 m above the CoG → -0.10 m. Pre-4.6: GPS_POS1_Z

AP-RTK X20D Firmware

• Available Firmware

Type	Version	Date
AP_Periph (DroneCAN update)	1.0.6	2026-09-22
AP_Periph (.apj package)	1.0.6	2026-09-22
Bootloader + App (merged HEX)	1.0.6	2026-09-22
Release verification (JSON)	1.0.6	2026-09-22

Download files from the product page (QR).

AP-RTK X20D Safety Notes

- Always remove propellers while wiring, configuring or updating firmware.
- Pull connectors by the housing, never by the wires.
- Beware of static discharge; do not operate in wet or dusty conditions.
- Firmware 1.0.1 is an engineering release — do not fly before first-article hardware and flight qualification.
- Antenna direction is opposite to AP-RTK dual (ANT1 rear, ANT2 front); swapped cables give a 180° heading error.
- Board ID 6205 only — never flash dual (1085) or G5H (6206) images.

AP-RTK X20D Safety Notes



<Detailed Manual>

